

Linear Regression

Simple Definitions · Diagrams · Formulas · Easy Explanations

1. What Is Linear Regression?

Simple definition: Linear Regression is a supervised machine learning technique that finds the best straight line describing the relationship between an input (x) and an output (y), so we can predict y for new values of x.

Example: Predict a person's Salary (output) from their Age (input).

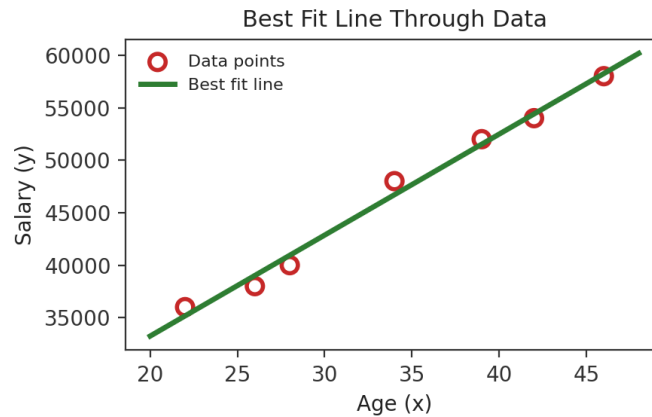


Fig 1: Data points and the best fit line through them

2. The Line Equation: $y = mx + c$

$$y = mx + c$$

Symbol	Meaning
x	Input data point
y	Predicted output
m	Slope - how steep the line is
c	Intercept - value of y when x = 0

3. Residual Error and the Cost Function

Residual error = Actual value - Predicted value, for each point.

Cost function = the **average** of all these residual errors (squared), across every data point:

$$\text{Cost} = (1 / 2n) \sum [(mx^{(i)} + c) - y^{(i)}]^2$$

- The **lower** the cost, the **better** the predictions.
- The cost depends entirely on the line we draw - which is decided by **m** and **c**. Change m or c → the line changes → the cost changes.

4. Gradient Descent

Simple definition: You try out **different lines** (different slope values), work out the **cost function for each one**, and plot all these costs. Then you simply **choose the line with the minimum cost**. That's Gradient Descent in action.

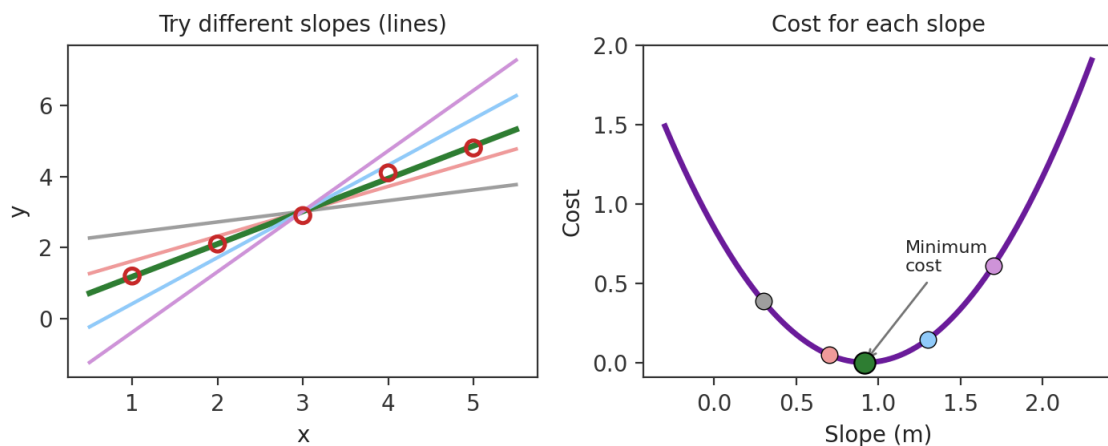


Fig 2: Left - several candidate lines (different slopes) through the data. Right - the cost of each line plotted against its slope; the lowest point is the best line.

- **Learning Rate (α):** Controls how big each step is while searching for the minimum cost. Too small → slow. Too large → may overshoot.
- **Convergence:** The point where the cost stops decreasing - we've found (close to) the best m and c.

5. Multiple Linear Regression

Real-world predictions often depend on **more than one input**. Instead of a single feature x , we now have **different features** $x_1, x_2, x_3 \dots$ and **one output** y (e.g. predicting Salary from Age, Location and Sleep hours):

$$y = m_1x_1 + m_2x_2 + m_3x_3 + \dots + m_nx_n + c$$

Each x is a different input feature, and each m is the weight/importance given to that feature. With two features the "best fit line" becomes a **best fit plane**, and with more features it becomes a higher-dimensional surface - but the idea stays the same: many inputs, one predicted output.

6. Performance Metrics: R^2 and Adjusted R^2

R^2 tells us how well the line explains the variation in the data (closer to 100% = better). **Problem:** R^2 keeps rising whenever we add any new feature, even a useless one. **Adjusted R^2** fixes this by penalizing features that don't help:

Features	R^2	Adjusted R^2	Trend
1	81%	79%	-
2	88%	89%	improved
3 (useless feature)	89%	86%	dropped

7. Overfitting vs. Underfitting

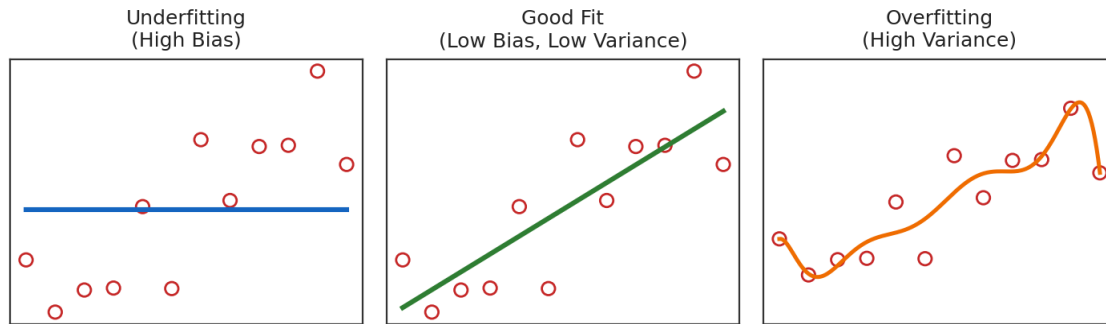


Fig 3: Underfitting, a good fit, and overfitting side by side

Model	Training	Testing	Result
M1	92%	64%	Overfitting (High Variance)
M2	92%	90%	Perfect Model
M3	68%	64%	Underfitting (High Bias)

M2 is ideal - it performs almost equally well on training and testing data.

8. Ridge and Lasso Regression

Simple definition: Ridge and Lasso are improved versions of Linear Regression that prevent overfitting by adding a **penalty term** to the cost function, discouraging very large slope values. This is called **Regularization**.

$$\text{Cost} = (1 / 2n) \sum [(mx^{(i)} + c) - y^{(i)}]^2 + \lambda(\text{slope})^2$$

- λ (**lambda**) = regularization strength - higher λ shrinks the slope more.
- **Ridge** squares the slope in the penalty - shrinks weights close to zero.
- **Lasso** uses the absolute value - can shrink some weights all the way to zero, removing less useful features.

Summary: Linear Regression finds the best line/plane by minimizing a cost function - gradient descent tries different lines and keeps the one with minimum cost. Models are evaluated with R^2 /Adjusted R^2 , checked for overfitting/underfitting, and can be regularized using Ridge or Lasso.